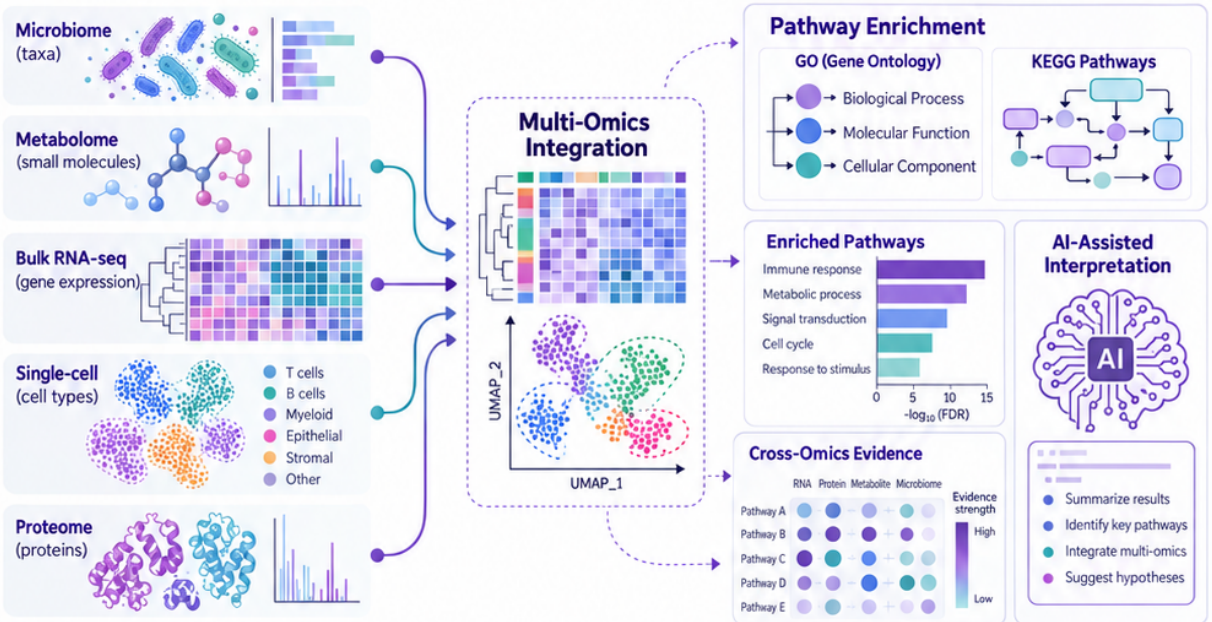
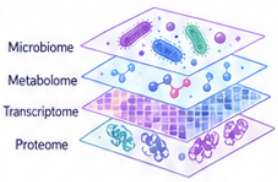
BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 9 READING MATERIAL

Multi-Omics Integration and Pathway Enrichment

From separate omics results to shared biological axes, pathway evidence, and AI-assisted interpretation





MULTI-OMICS INTEGRATION

Connect transcriptome, microbiome, metabolome, and proteome data to discover shared biological patterns.

PATHWAY ENRICHMENT

Use GO, KEGG, ORA, and GSEA to interpret gene lists and ranked statistics.

EVIDENCE SYNTHESIS

Combine pathway-level signals across layers to build defensible biological hypotheses.

AI-ASSISTED ANALYSIS

Leverage AI to organize workflows, verify outputs, and support scientific interpretation.

Fall 2026 | Two 40-Minute Undergraduate Sessions | Dr. Liwei Xie

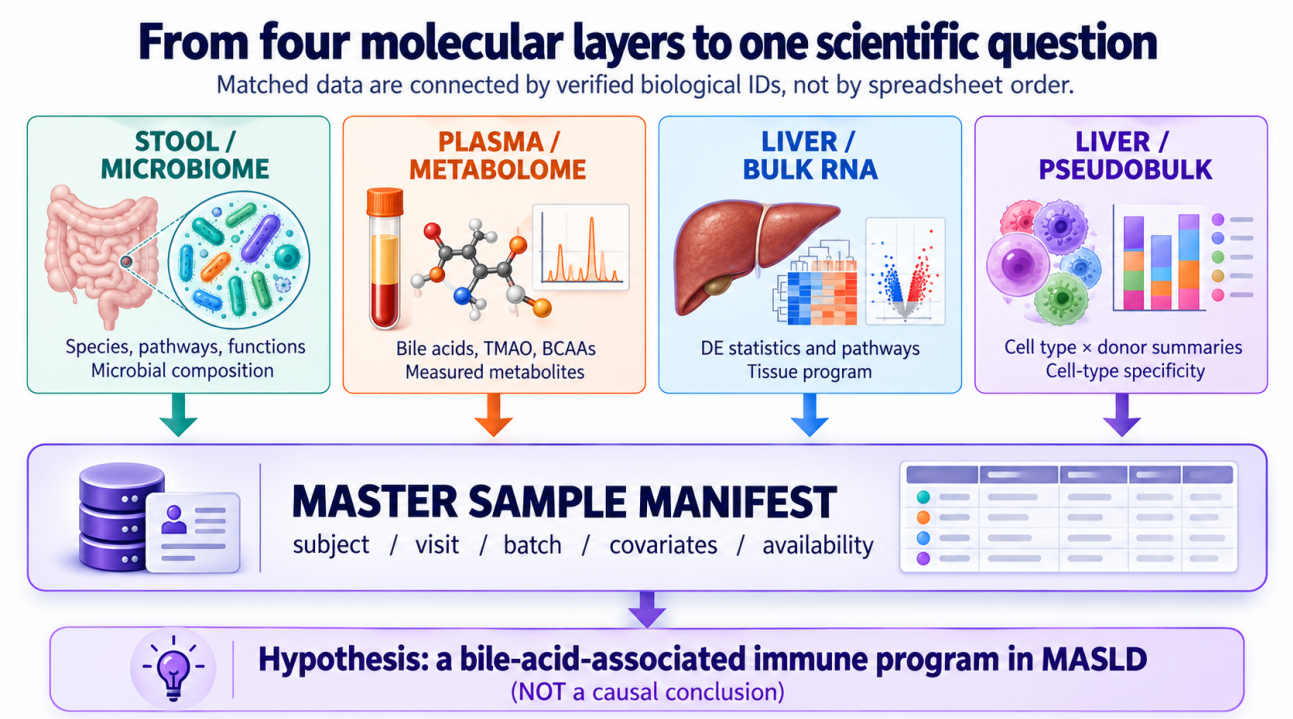


Figure 1. Conceptual study design. Four molecular assays enter a sample-level audit before any integrative model; the MASLD example is illustrative and contains no real patient estimates.

> *KEY TAKEAWAY Integrate biological evidence—not merely matrices or attractive heatmaps. Association, prediction and causation answer different questions.*

How to use this reading material

Week 9 is the synthesis point of Weeks 4–8: bulk and single-cell transcriptomics, microbiome profiling, proteomics/metabolomics, and spatial data. Session 1 explains when and how molecular layers should be integrated; Session 2 converts gene-level results into defensible pathway claims. The running case is a hypothetical gut–liver MASLD study with matched stool microbiome profiles, plasma metabolomics, and liver bulk plus cell-type pseudobulk RNA-seq. All schematic plots are teaching illustrations, not experimental findings.

Students do not need to write an R program from memory. They must be able to specify an analysis question, tell an AI assistant what assumptions to check, examine the returned code, detect invalid biological interpretations, and reproduce key outputs. Code blocks below are templates that require correctly formatted local input files and package installation; they are not claims that the illustrative results have been executed.

Learning outcomes

- Describe early, intermediate and late integration and select a strategy from sample matching, task, and assay geometry.
- Explain MOFA2 factor scores/weights and the predictive goal of DIABLO, including nested validation.
- Perform and audit GO/KEGG/Reactome ORA and preranked GSEA, with appropriate gene universe, ID mapping, FDR and gene-set versions.

- Build a multi-omics pathway-evidence table without combining incompatible effect scales or implying causality.
- Evaluate contemporary multimodal models and AI agents using reproducibility, transfer and scientific-integrity checks.

SESSION 1 | From separate omics layers to shared biological axes

1.1 Scientific design comes before integration

An integrated analysis needs a unit of observation and a specific inferential target. For clinical bulk datasets, the independent unit is commonly the subject or donor, not the sequencing library, repeated visit, cell, or tissue spot. Align samples by subject_id + visit/timepoint + biospecimen mapping, document cross-assay availability, and do not silently collapse repeated observations. In single-cell pseudobulk, summarize raw counts by donor × cell type (and condition/visit where relevant), then model donor-level replicates; individual cells should not be treated as independent biological replicates.

Omics layers differ fundamentally. RNA-seq has count-derived statistics and many genes; metabolomics has heterogeneous measurement scales and incomplete compound identification; microbiome relative abundances are compositional, sparse, and strongly affected by library size and prevalence; proteins may be censored by detection limits. Layer-specific preprocessing must precede any shared model. Sample-size adequacy, covariates, technical batches, and meaningful missingness are research decisions, not optional housekeeping.

Audit item	Minimum report	Why it matters
Sample matching	Unique donor + visit key; N per layer and intersection	False matches manufacture cross-layer covariance
Covariates	Disease definition, age, sex, medication, BMI, site	Confounding can dominate latent factors
Batch/protocol	Extraction, instrument run, library chemistry	A batch factor is not a biological program
Transform/filter	RNA VST/logCPM; metabolite log/scale; compositional-aware microbiome	Distorted feature scales and zero handling
Missingness	Sample and feature proportions, plus missingness mechanism	Unjustified imputation can invent signal
Feature IDs	Ensembl/NCBI/HMDB/KEGG, organism, release	Wrong mapping changes downstream pathways

Minimum sample manifest fields: sample_id, subject_id, visit, tissue, disease_state, age, sex, medications, site, collection_batch, extraction_batch, platform_batch, RNA_available,

metabolite_available, microbiome_available and scRNA_available. File checksums, preprocessing scripts and exclusion reasons belong in the project record.

1.2 Early, intermediate, and late integration

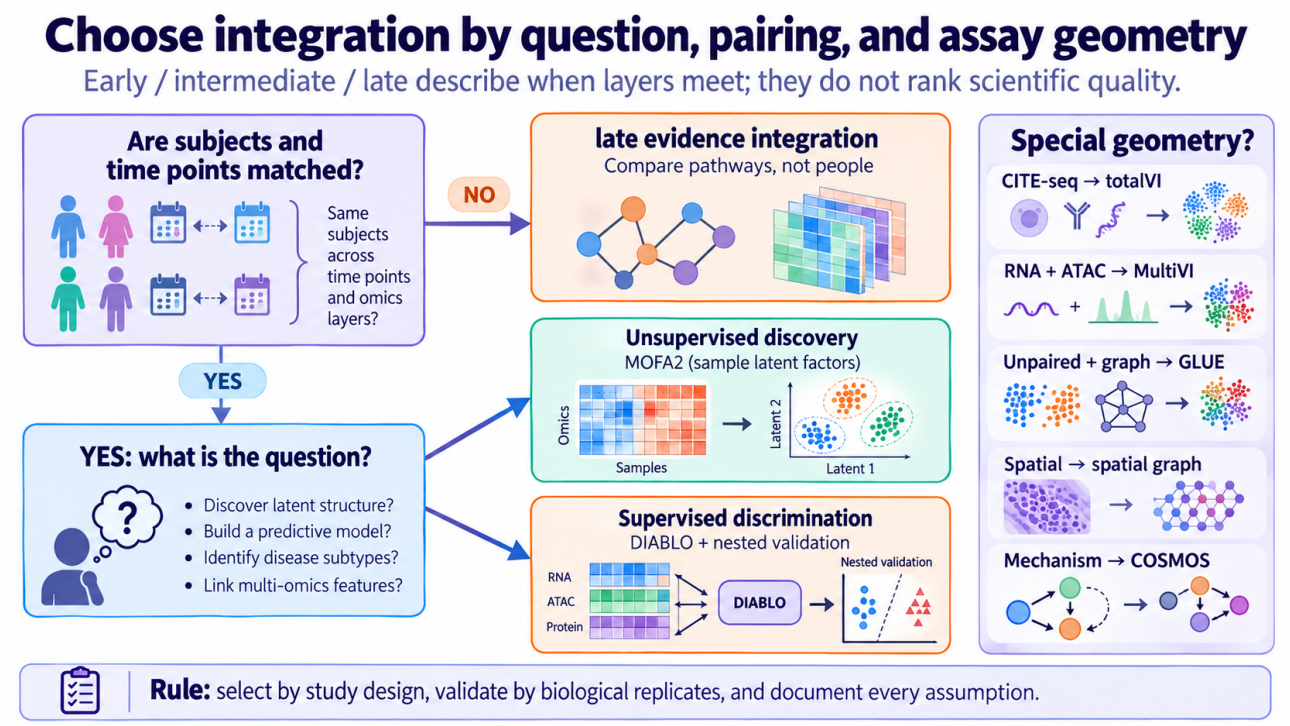


Figure 2. Method-selection tree. Pairing, scientific goal and experimental technology determine which integration method is defensible. Specialized single-cell and spatial methods solve different geometry problems.

Strategy	What is joined	When useful	Primary failure mode
Early	Scaled features before fitting	Carefully matched, well-powered cohorts	Block dominance; overfitting
Intermediate	Latent factors or multi-block components	Matched/overlapping samples; discovery or prediction	Factors may track batch; unstable selection
Late	Evidence after layer-specific analyses	Partial overlaps, smaller cohorts, pathway synthesis	Covariance is not modeled directly

Decision exercise. For 30 matched donors without labels, MOFA2 is a sensible discovery model; DIABLO is not appropriate without a well-defined supervised outcome. For matched labelled donors and a credible validation plan, DIABLO can identify a cross-omics classification signature. If assays represent disjoint people, pathway comparison can generate hypotheses, but it cannot establish within-person molecular relationships. A sample count of 20–30 is not sufficient by itself to justify a high-dimensional diagnostic classifier.

1.3 MOFA2: what is a latent factor?

MOFA2 learns lower-dimensional sources of variation across multiple views (Argelaguet et al., 2018, 2020). A factor score locates a sample along an inferred axis; a feature weight describes a feature's contribution relative to factor orientation; variance explained indicates how strongly the factor represents each view. Factors can be shared, view-specific, biological or technical. Their signs are arbitrary up to global factor flipping, and the magnitude of a weight is not a causal effect size. Post hoc association with MASLD phenotype is a separate statistical question.

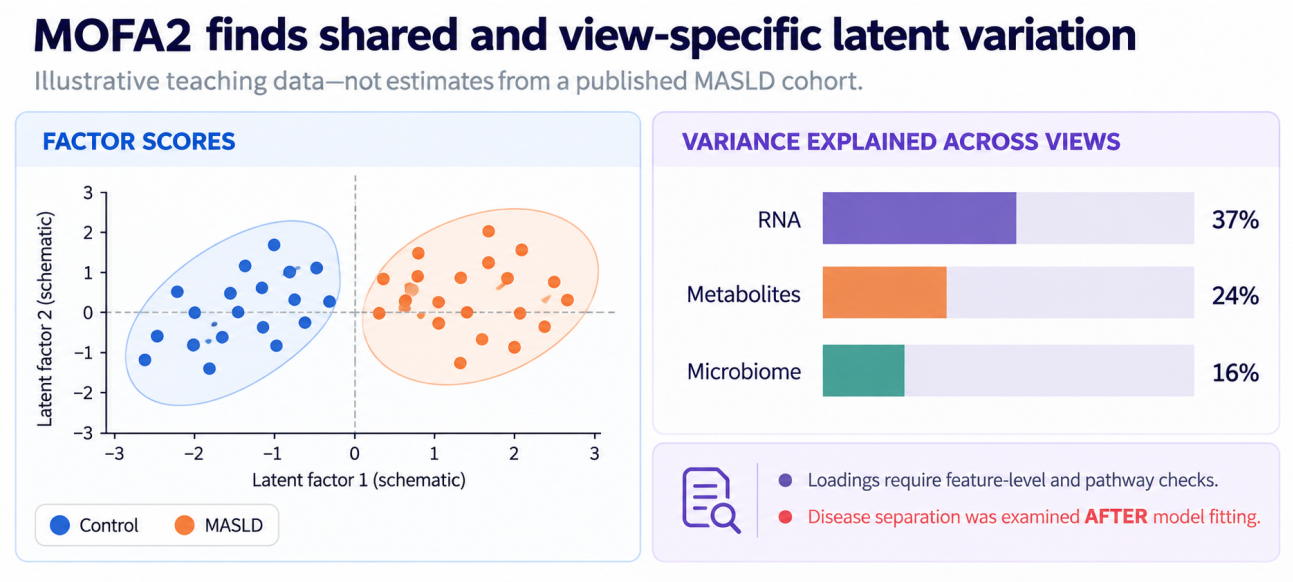


Figure 3. Simulated factor plot and variance-explained bars. A disease-associated factor is still an observational axis; its validity depends on variance across views, confounder checks and stable loadings.

Output	Interpretation	Required check
Factor score	Sample coordinate	Correlate with biological AND technical metadata
Feature weight	Contribution to a factor	Check sign convention, resampling stability, pathways
Explained variance	Variance reconstructed per view	Confirm that purported shared factors are shared
Missing view	Not all blocks observed	Inspect missingness assumptions and sensitivity

Robust preprocessing principle. Remove constant features before z-scoring; otherwise scale() can create NA values. Inspect original missingness before filtering. If a view is absent for a donor, MOFA2 may support a missing view, but this does not mean missingness is random. In a teaching exercise, a fully matched intersection is easier to audit; in a study, explain why the selected design is appropriate.

R workflow 1 | MOFA2 audit + factor fitting

```
library(MOFA2)
library(dplyr)
# Matrices: feature x sample; already quality-filtered / transformed.
read_view <- function(f) as.matrix(read.csv(f, row.names=1, check.names=FALSE))
rna <- read_view("rna_vst.csv")
met <- read_view("metabolites_log.csv")
mic <- read_view("microbiome_clr.csv")
meta <- read.csv("metadata.csv", check.names=FALSE)
stopifnot(!anyDuplicated(meta$sample_id))
ids <- meta$sample_id[meta$sample_id %in% Reduce(intersect,
          list(colnames(rna),colnames(met),colnames(mic)))]
stopifnot(length(ids)>=10)
message("Fully matched donors: ", length(ids))
zfeat <- function(z) {
 z <- z[,ids,drop=FALSE]; keep <- apply(z,1,function(v)
          all(is.finite(v)) && sd(v)>0)
 message("Rows retained: ",sum(keep)," / ",nrow(z))
 t(scale(t(z[keep,,drop=FALSE])))
}
views <- list(RNA=zfeat(rna),Metabolites=zfeat(met),Microbiome=zfeat(mic))
stopifnot(all(vapply(views, function(z) all(is.finite(z)),logical(1))))
m <- create_mofa(views)
m <- set_data_options(m, scale_views=FALSE)
m <- set_model_options(m, num_factors=min(8L,length(ids)-1L))
m <- set_train_options(m,seed=2026,convergence_mode="slow",
                  outfile="mofa_week9.hdf5")
m <- run_mofa(m)
print(get_variance_explained(m)$r2_per_factor)
weights <- get_weights(m, factors=1, as.data.frame=TRUE)
write.csv(weights,"mofa_factor1_weights.csv",row.names=FALSE)
# Next: inspect model convergence; join factor scores with metadata
# and test associations without calling them causal mechanisms.
```

AI/agent prompt. “Write an R audit for matched RNA, metabolite and microbiome matrices before fitting MOFA2. Print all sample mismatches, constant-feature exclusions, missingness by view, explained variance per factor and correlations with batches. Never use case/control to fit factors. Refuse to interpret an axis until these checks are visible.”

1.4 DIABLO: supervised multi-block signatures

DIABLO (mixOmics) selects correlated cross-block features that discriminate prespecified groups (Singh et al., 2019). Model performance is a property of predictions on unseen subjects, not the visual separation of training samples. The design matrix describes which blocks are encouraged to covary; keepX selects features per block and component; ncomp determines complexity. Every adaptive choice must be inside the training portion of a validation design.

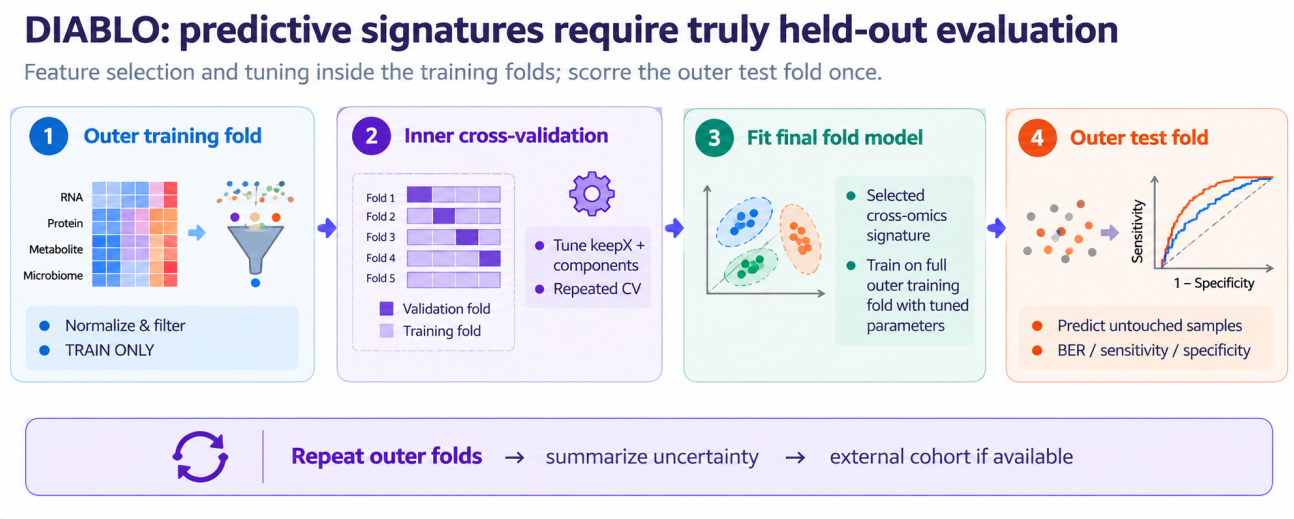


Figure 4. Nested-validation logic. Use outer held-out donors to estimate generalization while inner folds tune keepX and component count. Preserve grouping of repeat visits and related specimens.

Critical distinction: repeated CV used both to select and report the best configuration is not an unbiased performance estimate. Nested cross-validation or a locked external test set is preferred. At low n, report model instability and confidence intervals; a clinical-biomarker claim requires substantially stronger validation.

R workflow 2 | DIABLO nested-validation skeleton

```
library(mixOmics)
# Input: same subjects, rows=samples, cols=features; no missing values.
X <- list(RNA=t(views$RNA),Metabolites=t(views$Metabolites),
          Microbiome=t(views$Microbiome))
y <- factor(meta$condition[match(ids,meta$sample_id)])
stopifnot(all(vapply(X,function(z) identical(rownames(z),ids),logical(1))))
stopifnot(nlevels(y)==2 && min(table(y))>=5)
set.seed(2026)
outer <- sample(rep(1:5,length.out=length(ids)))  # teaching only;
# For real data: subject-grouped + class-stratified outer folds.
outer_predictions <- data.frame()
for (fold in 1:5) {
 itr <- which(outer!=fold); ite <- which(outer==fold)
 Xt <- lapply(X,function(z) z[itr,,drop=FALSE])
 Xv <- lapply(X,function(z) z[ite,,drop=FALSE])
 yy <- y[itr]
 if (min(table(yy))<3) next
 design <- matrix(.1,3,3,dimnames=list(names(X),names(X)))
 diag(design) <- 0
 tuned <- tune.block.splsda(Xt,yy,ncomp=2,design=design,
          test.keepX=list(RNA=c(10,20),Metabolites=c(5,10),
                          Microbiome=c(5,10)),
          validation="Mfold",folds=3,nrepeat=5,
          dist="centroids.dist",measure="BER",progressBar=FALSE)
 fit <- block.splsda(Xt,yy,ncomp=tuned$choice.ncomp$BER,
```

```
                keepX=tuned$choice.keepX,design=design)
 pred <- predict(fit,newdata=Xv,dist="centroids.dist")
 # Check returned prediction array shape in installed mixOmics version;
 # save held-out labels and predicted classes for BER calculation.
 message("Outer fold ",fold,": train ",length(itr),"; test ",length(ite))
}
# Refit on the full development cohort only after finalizing design.
# Report outer-fold BER, class sensitivity and stability, not train accuracy.
```

Instructor note. The outer-loop code demonstrates information boundaries, not a turnkey clinical benchmark. Real studies must implement stratified/grouped splits, training-fold preprocessing and explicit extraction of held-out class predictions under their installed mixOmics version. Ask the agent to provide the resulting prediction schema and unit tests before accepting scores.

1.5 Frontier methods: 2025–2026 progress and limits

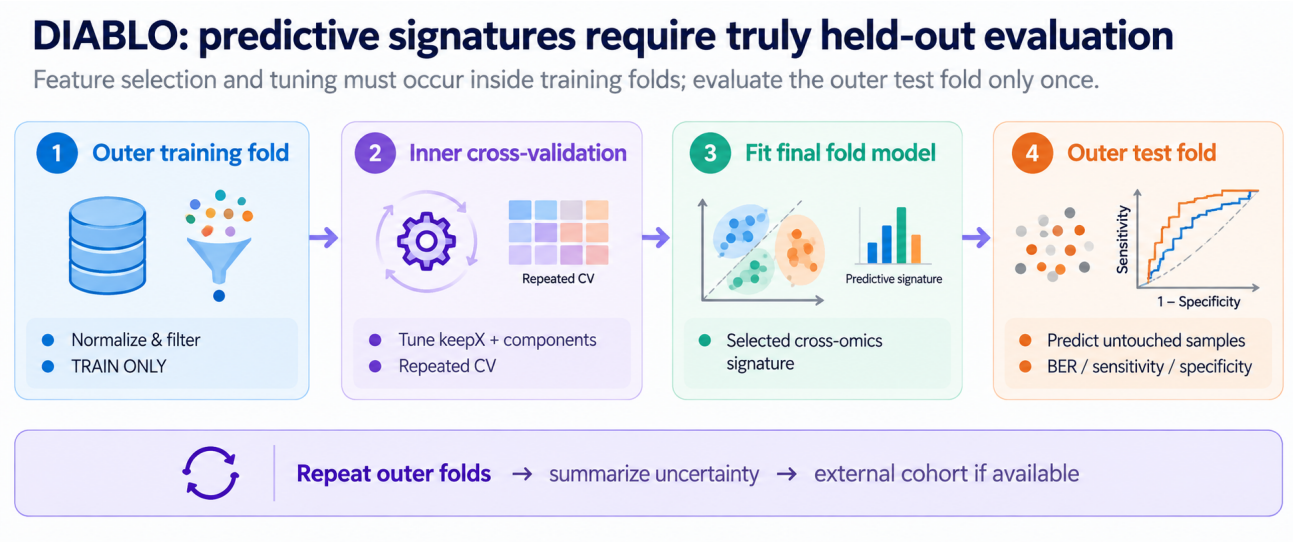


Figure 5. Method families and their principal validation hazards. Recent 2025–2026 benchmarks emphasize task- and dataset-specific performance rather than a universal model ranking.

Two independent 2025 Nature Methods benchmarking papers evaluated around 40 single-cell multimodal integration methods across integration settings and downstream tasks. A 2026 Nature Communications study (SCMBench) assessed 23 domain-specific and foundation-model approaches and highlighted the importance of biomarker recovery, trajectories, and batch correction, beyond visual embedding quality [15–17]. These are external research developments, not results from the hypothetical MASLD example.

Assay/question	Method family	What to validate
CITE-seq counts	totalVI	Protein-background model, held-out donors, batch

RNA + ATAC / partial mosaic	MultiVI	Modality reconstruction and relevant cell populations
Unpaired scRNA + scATAC	GLUE	Regulatory prior and correspondence accuracy
Spatial multimodal neighborhoods	SpatialGlue / spatial GNNs	Section-level holdouts; spatial autocorrelation
Phospho → RNA → metabolite	COSMOS + prior networks	Sensitivity to graph/priors; experimental intervention
Reference/query cells	scGPT, Geneformer and related foundation models	OOD populations, rare-cell recall, uncertainty

What changes from the original Week 9? Foundation models are presented as pretrained representation tools, not validated causal interpreters. Cross-dataset transfer, cellular composition shifts, reference imbalance and prior-graph quality are explicit evaluation targets. A model that aligns two UMAPs beautifully can still be biologically incorrect; independent marker biology and held-out donors remain essential.

SESSION 2 | From genes to pathways and testable biological hypotheses

2.1 GO, KEGG, Reactome and gene-set collections

Gene Ontology separates Biological Process (BP), Molecular Function (MF) and Cellular Component (CC); GO annotations differ in evidence provenance. The GO 2026 update introduced revised annotations and expanded GO-CAM knowledge, including structured models of pathways that describe relations among activities rather than just gene lists [8]. KEGG provides molecular interaction and reaction pathway maps [9]; Reactome offers curated reaction/event hierarchies and updated multi-omics exploration in its 2026 knowledgebase [10]. MSigDB Hallmark collections reduce redundancy relative to many overlapping GO terms, but should be versioned like any other database.

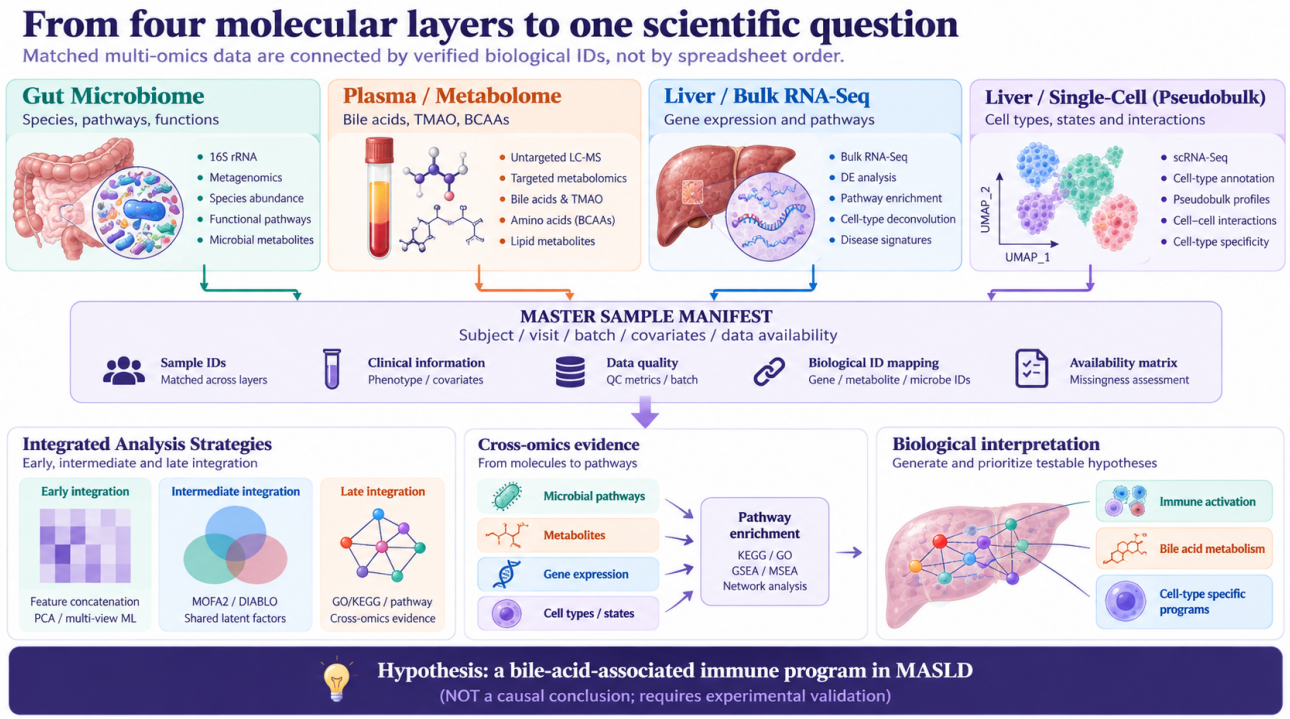


Figure 6. Four functional resources and one non-negotiable requirement: annotation provenance. GO terms are not the same objects as KEGG metabolic maps or curated Reactome reactions.

ORA and preranked GSEA describe enrichment relative to a specific null; they do not measure pathway flux or enzyme activity. A positive GSEA normalized enrichment score (NES) means the set is enriched near the positive end of the ranking metric, whose group contrast must be stated. For highly redundant term sets, clusterProfiler simplify() (GO) or semantic clustering can improve display without changing the underlying research question.

Tool/resource	Primary object	Use / caution
GO BP/MF/CC	Ontology-term annotation	Function/localization; term redundancy
GO-CAM	Activity-causal graph model	Curated mechanistic knowledge, not experimental causality
KEGG	Organism-specific pathway map	Compound/gene identity and organism version
Reactome	Manually curated reaction/event	Hierarchies and reaction direction may matter
MSigDB Hallmark	Curated expression-related gene signatures	Good first pass, not all pathways
MetaboAnalyst 6.0	Compound and integrated pathway tool	MSI confidence, identifier coverage, units

2.2 ORA versus preranked GSEA: what each statistical question means

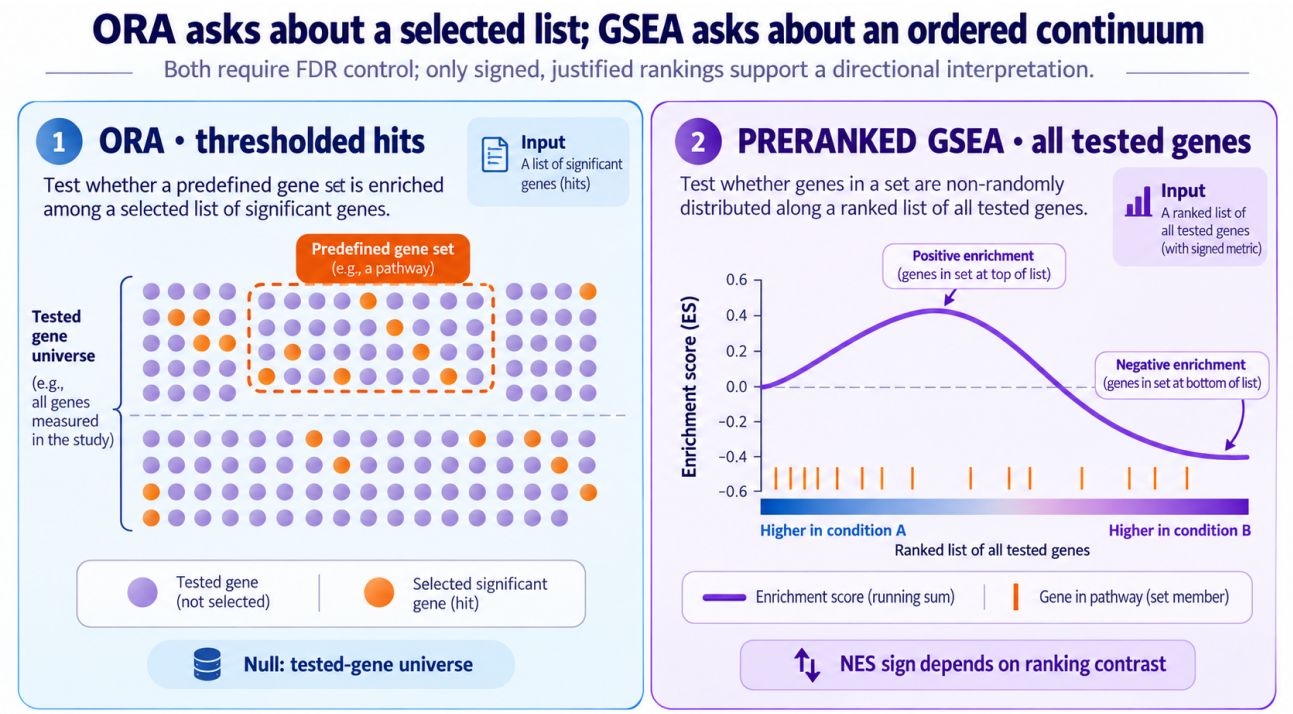


Figure 7. Two complementary designs. ORA uses thresholded genes and a defensible tested background; preranked GSEA uses a complete signed ranking and reports NES, FDR and leading-edge genes.

Over-representation analysis (ORA) uses the hypergeometric/Fisher framework to test whether a threshold-selected gene list intersects a pathway more often than expected. The background must be the set of genes eligible for selection (typically expressed and statistically tested), not all genes encoded in the genome. ORA is sensitive to DE cutoff, gene-detection bias, and gene-set size. A completely empty result is scientifically possible and should trigger mapping/universe checks rather than automatic loosening of cutoffs.

Preranked GSEA evaluates whether the members of a pathway accumulate at one extreme of a signed statistic, often the DESeq2 Wald statistic, including nonsignificant genes. The normalized enrichment score adjusts for set size. Leading-edge genes contribute to the extreme running-sum score, but their selection is descriptive. Gene–gene correlation and confounded rank structure can inflate evidence in some settings; methods such as CAMERA and sensitivity analyses help check robustness. GO enrichment results must be interpreted with attention to annotation evidence and biological direction.

> *KEY TAKEAWAY Negative NES does not mean an inhibited biochemical pathway. It means pathway genes are concentrated at the negative end of the stated gene-ranking contrast.*

2.3 An auditable clusterProfiler workflow in R

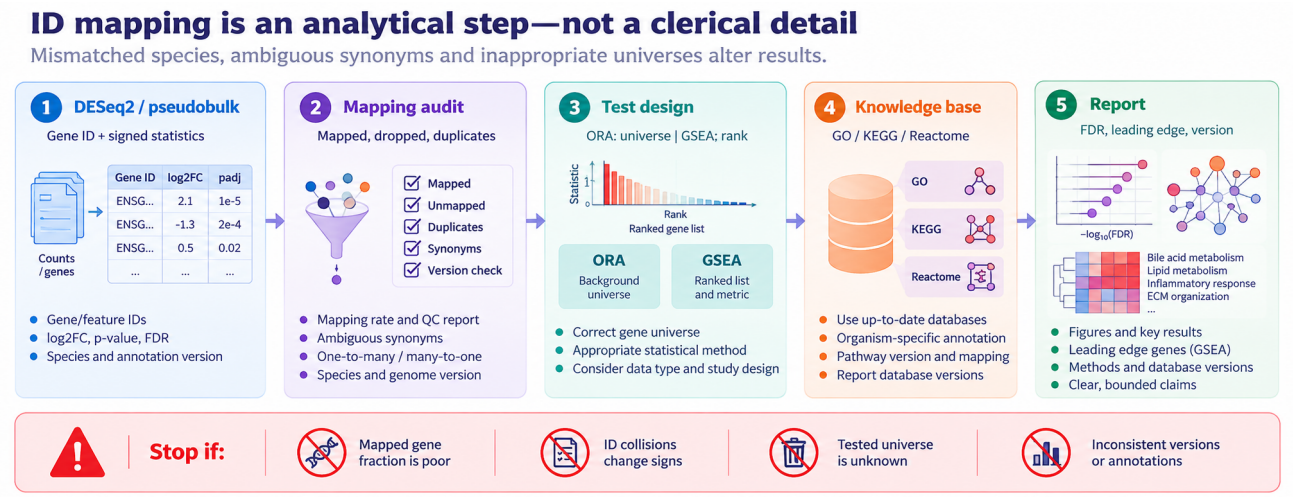


Figure 8. The ID and analysis audit. Species, gene sets, duplicate ID policy, FDR family and version are part of the reported analysis, not invisible software defaults.

Identifier hygiene. If a gene symbol maps to multiple Entrez IDs, the number of mapping rows is not the mapping success rate. Count uniquely mapped source genes and log every one-to-many relationship. When multiple original gene features collapse onto one Entrez ID, selecting the maximum signed statistic can reverse the intended biological direction: use the largest absolute statistic with its original sign, preferably after cleaning ambiguous gene IDs. Where possible, start from stable Ensembl gene IDs and record conversion tables.

R workflow 3 | ORA + GSEA with protected direction and gene universe

```
library(clusterProfiler)
library(org.Hs.eg.db)
library(dplyr)
library(enrichplot)
res <- read.csv("pseudobulk_DE_results.csv")
# Required: gene (HGNC symbol), stat, log2FoldChange, padj.
stopifnot(all(c("gene","stat","log2FoldChange","padj") %in% names(res)))
res <- res %>% filter(!is.na(gene),nzchar(gene))
U <- unique(readLines("tested_gene_universe.txt"))
ids <- unique(c(U,res$gene))
map <- bitr(ids,fromType="SYMBOL",toType="ENTREZID",OrgDb=org.Hs.eg.db)
cat("Unique SYMBOL mapping fraction:",
    length(intersect(U,unique(map$SYMBOL)))/length(U),"
")
# Keep only statistically tested genes as the ORA universe.
uni <- unique(map$ENTREZID[map$SYMBOL %in% U])
sig <- res %>% filter(!is.na(padj),padj<.05,
                      abs(log2FoldChange)>1) %>% pull(gene) %>% unique()
hits <- unique(map$ENTREZID[map$SYMBOL %in% sig])
hits <- intersect(hits,uni)
stopifnot(length(uni)>0,length(hits)>0)
ora_go <- enrichGO(hits,OrgDb=org.Hs.eg.db,keyType="ENTREZID",
```

```
      ont="BP",universe=uni,pAdjustMethod="BH",readable=TRUE)
ora_kegg <- enrichKEGG(hits,organism="hsa",universe=uni,
                       pAdjustMethod="BH")
# Keep original sign when multiple SYMBOLs collapse to one Entrez ID.
ranked <- res %>% filter(is.finite(stat),gene %in% U) %>%
  inner_join(map,by=c("gene"="SYMBOL")) %>%
  arrange(desc(abs(stat)),gene) %>%
  distinct(ENTREZID,.keep_all=TRUE) %>% arrange(desc(stat),ENTREZID)
geneList <- ranked$stat;names(geneList) <- ranked$ENTREZID
stopifnot(!anyDuplicated(names(geneList)),all(is.finite(geneList)))
gsea_go <- gseGO(geneList,OrgDb=org.Hs.eg.db,keyType="ENTREZID",
         ont="BP",minGSSize=10,maxGSSize=500,eps=0,verbose=FALSE)
gsea_kegg <- gseKEGG(geneList,organism="hsa",minGSSize=10,
                     maxGSSize=500,eps=0,verbose=FALSE)
write.csv(as.data.frame(ora_go),"GO_BP_ORA.csv",row.names=FALSE)
write.csv(as.data.frame(ora_kegg),"KEGG_ORA.csv",row.names=FALSE)
write.csv(as.data.frame(gsea_go),"GO_BP_GSEA.csv",row.names=FALSE)
write.csv(as.data.frame(gsea_kegg),"KEGG_GSEA.csv",row.names=FALSE)
write.csv(map,"gene_identifier_mapping.csv",row.names=FALSE)
writeLines(capture.output(sessionInfo()),"R_sessionInfo.txt")
```

Additional required checks. Inspect the mapping collision report and gene-set membership; for GO BP/MF/CC run separate ontologies or ont="ALL" with an explicit pooling rule. Generate dotplot(), gseaplot2(), cnetplot() or a category–gene network only when result objects are nonempty. The full all-gene ranking must have a stable contrast, species code and signed statistic. Report KEGG data access date and GO/Reactome/MSigDB release. Some KEGG interfaces require live access, so a stable archived gene-set snapshot is preferable for exact reanalysis.

AI/agent prompt. “Inspect my DESeq2 results and tested-universe file, report unique mapping fraction and one-to-many mappings, define the class contrast, and implement ORA and preranked GSEA using valid input IDs and BH-FDR. If a requested package call is unavailable, consult its installed help and explain the change; do not invent function arguments or silently replace missing results.”

2.4 Integrating genes, proteins, metabolites and microbial functions

An evidence-first pathway analysis starts with trustworthy results in each layer. RNA and protein abundance support associations with expression, not enzyme flux. A detected metabolite supports a measured chemical feature only to the extent its annotation is reliable; tentative isomers and generic lipid classes must not be mapped as specific KEGG compounds without appropriate evidence. Shotgun-metagenomic KO profiles estimate gene functional potential; 16S-based PICRUSt2 outputs are predictions, not measured microbial gene content or activity.

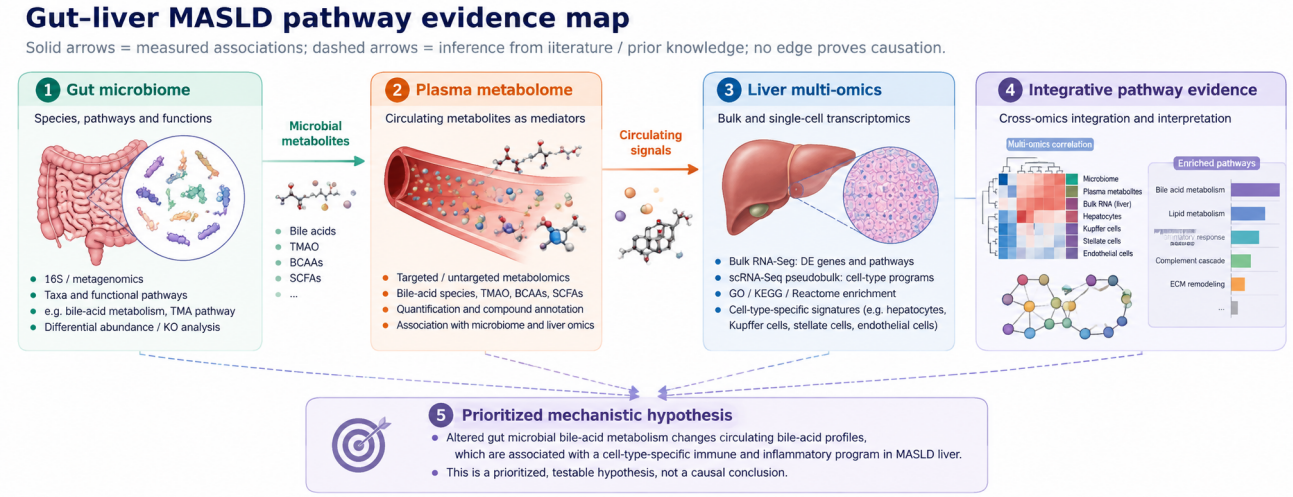


Figure 9. Case study hypothesis network. Dashed edges indicate inferred microbial potential or a literature hypothesis; solid edges represent explicitly measured associations. Edge arrows do not establish direction of causation.

Joint pathway tools such as MetaboAnalyst 6.0 can identify gene–compound pathway overlap after explicit mapping. ActivePathways combines gene-level p-values across evidence sources, but its input is gene-indexed, calibrated significance evidence—not arbitrary microbiome taxa and compound identifiers placed into one table. Analyses sharing donors or biological dependencies should not be described as independent replication. Distinguish orthogonal assay support from independent cohort replication.

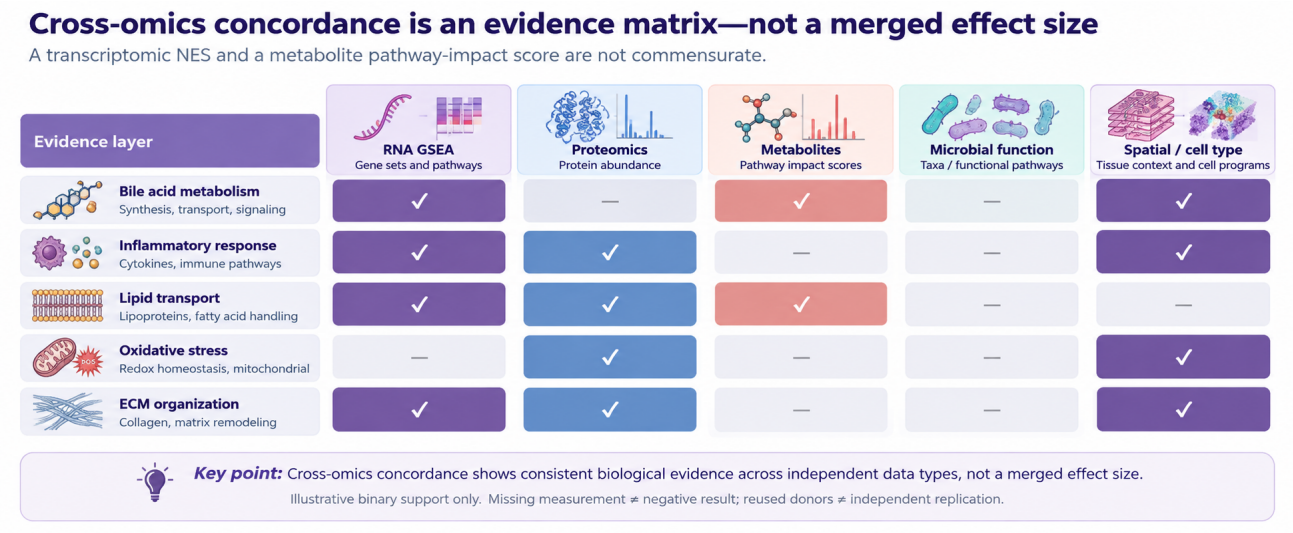


Figure 10. Illustrative evidence-support matrix. A binary evidence marker is suitable for summarization; quantitative effects require separate scales and units. Missing modalities are neither negative results nor zero effects.

Field	Example entry	Interpretation
Pathway ID / release	hsa04976 / KEGG date	Reproducible cross-reference
RNA result	NES, FDR, leading edge	Signed ranked-gene association

Metabolite result	Compound IDs, MSI level, FDR	Measured metabolite perturbation
Microbiome result	Taxon + KO measurement/prediction	Taxa/functional potential; composition-aware
Cellular context	Macrophage pseudobulk, donor N	Localization but not causality
Evidence grade	Association / convergence / replication	Report exact strength and limitations

R workflow 4 | Evidence table that keeps incomparable metrics separate

```
library(dplyr)
library(readr)
# Both files must use real, database-matched KEGG IDs; no name-only join.
rna <- read_csv("KEGG_GSEA.csv",show_col_types=FALSE) %>%
 transmute(pathway_id=ID, rna_name=Description,
          rna_NES=NES, rna_FDR=p.adjust)
met <- read_csv("metabolite_pathway_results.csv",show_col_types=FALSE) %>%
 transmute(pathway_id=kegg_pathway_id, met_name=pathway_name,
          met_FDR=fdr, met_impact=impact,
          compound_ids=matched_compound_ids)
stopifnot(!anyDuplicated(rna$pathway_id),!anyDuplicated(met$pathway_id))
evidence <- full_join(rna,met,by="pathway_id") %>%
 mutate(pathway=coalesce(rna_name,met_name),
       rna_support=!is.na(rna_FDR)&rna_FDR<0.05,
       met_support=!is.na(met_FDR)&met_FDR<0.05,
       convergent=rna_support&met_support,
       rna_rank_direction=case_when(is.na(rna_NES)~"not tested",
            rna_NES>0~"positive end",TRUE~"negative end"))
write_csv(evidence,"multiomics_pathway_evidence.csv")
# Plot a categorical support matrix, not RNA NES and met_impact
# in one shared numeric color scale.
print(evidence %>% count(rna_support,met_support))
```

Recommended manuscript sentence. “Liver donor-level pseudobulk GSEA showed FDR-controlled enrichment of genes associated with bile-acid transport in the prespecified MASLD contrast. Plasma bile-acid profiles showed complementary changes, and macrophage-associated inflammatory terms localized part of the transcriptomic pattern. These observations are consistent with a perturbed gut–liver bile-acid axis but do not establish upstream microbial causation.”

2.5 From a pathway result to an experiment

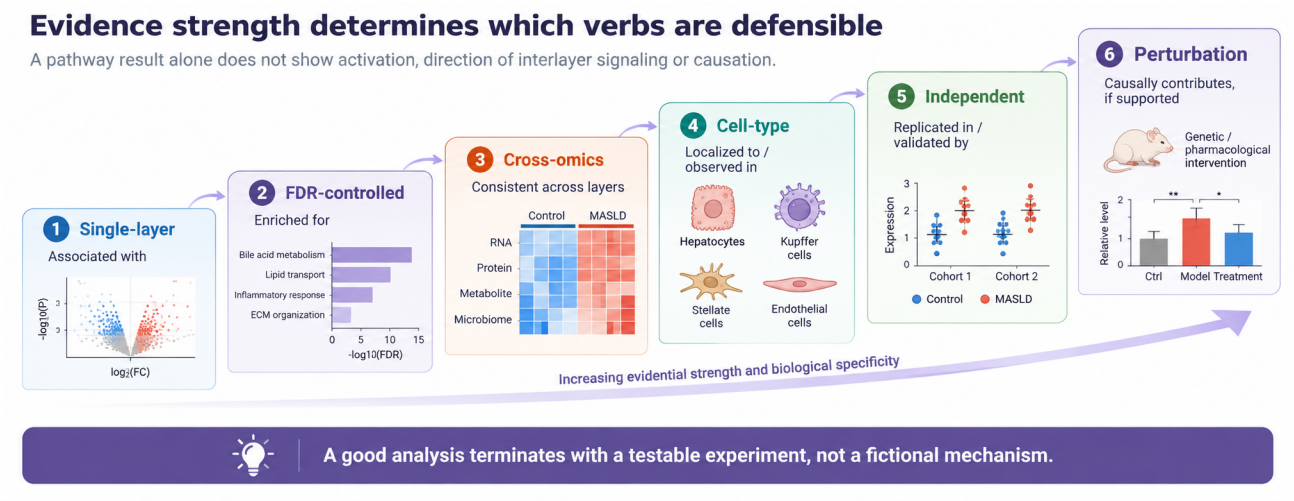


Figure 11. Evidence ladder. Each step justifies a stronger but still bounded description; perturbation findings need their own controls, timing and rescue experiments.

For the running case, ask a sequence of falsifiable questions: Is the plasma bile-acid signal associated with stool taxa after medication, diet and disease-stage adjustment? Is the liver pathway result reproducible across donors and gene-set collections? Does the macrophage pattern persist in cell-type pseudobulk after composition adjustment? Are metabolites identified at sufficient confidence to justify pathway mapping? Would microbiome depletion/transplantation, a candidate metabolite perturbation, or receptor-level rescue establish directionality? An observational pathway plot cannot answer the final question.

Useful alternatives. Adjusted association analysis, mediation models and causal-network software can help generate testable hypotheses, but observational mediation depends on strong causal assumptions. Time ordering, negative controls, independent cohorts, genetic or chemical perturbation, and rescue move the interpretation from “consistent with” toward “causally contributes to.”

2.6 AI assistants, agent workflows and scientific accountability

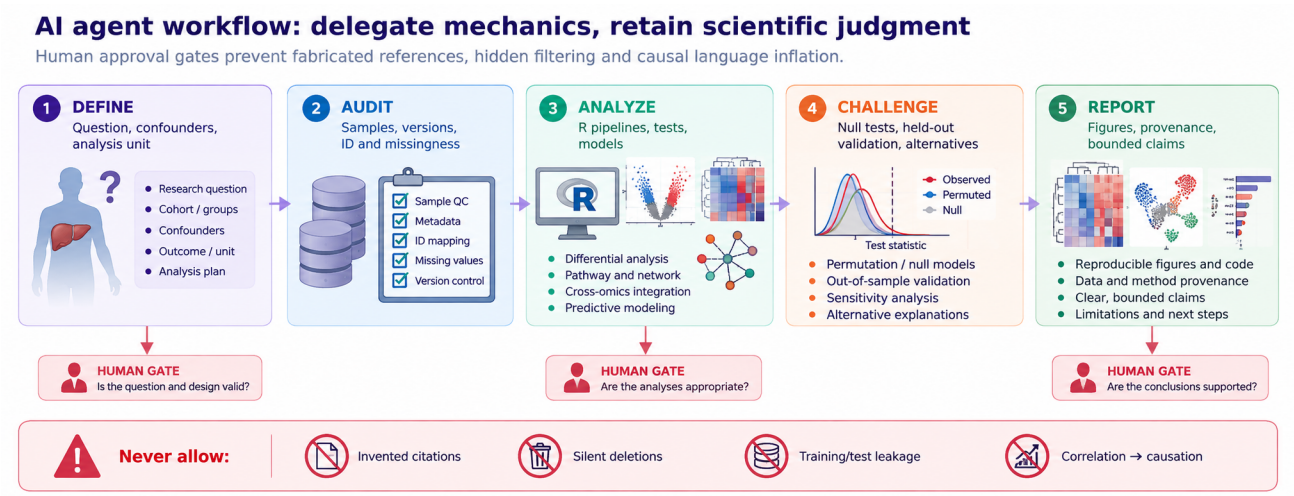


Figure 12. Human-in-the-loop agent structure. Scientific claims require deliberate approval gates even when automated code is syntactically valid.

AI agents are effective at producing R scaffolds, documentation, identifier-audit summaries and draft figures, but they can also choose the wrong organism, miss duplicate genes, impute without consent, tune on the test set, cite nonexistent papers and write causal conclusions from descriptive plots. "The code ran" is necessary but not sufficient evidence that the analysis answers the right biological question.

Agent stage	What AI may do	What the student must decide
Question	Suggest pipeline and alternative hypotheses	Define target, biological replicate, confounders
Audit	Build manifest, missingness and ID report	Approve exclusions, mapping and versions
Analysis	Generate R code and intermediate QC files	Inspect assumptions, effect sizes, nulls
Validation	Prepare nested CV/robustness scripts	Choose appropriate test and interpret limitations
Reporting	Render figures, figure legends, literature list	Check every statistic, citation and scientific verb

Reusable instruction to an AI agent. "You are an analysis assistant, not a scientific authority. Before coding, list the scientific question, input schemas, independent unit, model assumptions, covariates, planned QC, null hypothesis and acceptance criteria. Never change the sample set or thresholds silently. Produce a machine-readable audit, per-stage outputs, failed checks, full sessionInfo(), and a clearly marked interpretation restricted to measured evidence. Provide real DOI-backed citations or state that a citation is unverified."

Common errors: the 2026 troubleshooting checklist

Error	What is wrong	Corrective action
IDs look mapped, >100% rate	Counts join rows, not source genes	Use unique source IDs; log one-to-many cases
GSEA changes sign unexpectedly	max(stat) collapses duplicated IDs	Use absolute rank and preserve original sign
ORA tests all human genes	Unmeasured genes used as background	Use expressed and tested universe
Pathway "activated" from ORA	ORA has no direction	Report over-representation only
Metabolite impact = RNA NES	Different units/estimands	Separate metrics or use categorical evidence

DIABLO nearly perfect training plot	Feature selection and assessment overlap	Held-out nested/grouped CV
"Independent" modalities share patients	Orthogonal measurement, not replication	Reserve replication for held-out subjects
Predicted 16S functions called measured	Inferred KO is not metagenomic evidence	Explicitly label prediction
Single cells used as n	Pseudoreplication	Donor-level pseudobulk/statistics
Foundation model treated as truth	Out-of-distribution failure	Benchmark rare cells, donors and batches

Week 9 practice questions

1. Define early, intermediate and late integration. What is the safest first integration strategy for two omics cohorts with no overlapping donors?
2. Twenty donors have matched multi-omics data but no outcome labels. Which model would you use first, and which variables would you audit before fitting?
3. MOFA Factor 2 explains 27% of RNA variation but only 1% of metabolomic and microbiome variation. Write a justified one-sentence interpretation.
4. Describe how selection bias enters DIABLO when the same CV scores select model complexity and estimate generalization.
5. Contrast ORA with GSEA using a hypothetical 12,000-gene tested universe and a signed DESeq2 statistic.
6. A 100-gene significant set yields 118 mapping rows. Can you report a 118% conversion rate? What must be counted instead?
7. One Ensembl/Entrez ID maps to two symbols with Wald statistics −4 and +2. Explain why selecting max(stat) is misleading.
8. Explain why positive NES is not evidence of greater metabolic flux through a KEGG reaction.
9. RNA GSEA, plasma bile acids and 16S-predicted KO abundance all implicate bile-acid metabolism. Write a two-sentence evidence statement without causal overclaim.
10. Compare a totalVI analysis of CITE-seq, MultiVI analysis of RNA+ATAC, and spatial GNN analysis of tissue sections. Specify one held-out validation strategy per model.
11. Identify three decisions an AI agent must not make silently in a multi-omics study.
12. Propose a perturbation/rescue experiment that could distinguish a microbiome-driven bile-acid mechanism from a disease-associated but noncausal microbial signature.

Project 4: expected submission package

Submit (i) one sample manifest and exclusions record, (ii) a layer-specific QC summary, (iii) a method-selection justification, (iv) GO/KEGG or Reactome enrichment tables with tested universe and annotated genes, (v) an interpretable evidence-concordance matrix, (vi) two figures with clear legends, (vii) one bounded hypothesis plus discriminating validation experiment, and (viii) the exact

AI prompt log and code/session information. A high-quality negative or non-concordant result is valid when it is reported transparently.

Assessment dimension	Weight	Criteria
Scientific design and biological unit	25%	Correct pairing, covariates, assay limitations
Reproducible analysis and identifiers	25%	Complete scripts, mapping audit, gene universe, versions
Statistical and model validation	20%	FDR, sensitivity, held-out design when applicable
Biological interpretation	20%	Evidence grades, correct verbs, falsifiable model
AI integrity and documentation	10%	Transparent agent use, checked citations and outputs

Suggested reading: foundational papers and 2025–2026 updates

References below provide direct DOI links and were checked against publisher, PubMed or official database records during the October 2026 revision. Teaching synthesis and diagrams are newly authored; they are not presented as figures reproduced from these sources.

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Bridge to Week 10

Bring one final enrichment table and one well-audited integration result (MOFA factor, DIABLO held-out signature, targeted association or pathway-concordance matrix). In Week 10 we will turn these into publication-quality figures with ggplot2, enrichplot, ComplexHeatmap and network methods. The central goal remains unchanged: discovery begins with a careful scientific question and ends with a transparent, falsifiable claim.

EXTENDED LAB REFERENCE | Study independently after the two lectures

The following reference modules are not additional live-lecture requirements. They are optional self-paced reading, designed to help students work without prior R programming experience. Each module follows the same rule: articulate a scientific choice, delegate implementation to an AI assistant, and verify the result independently.

Appendix A | Sample identity, missingness and pseudoreplication

The most damaging integration error can be entirely invisible in a heatmap: two assays labeled as the same donor do not represent the same biological specimen or visit. Consider a longitudinal MASLD study where stool was collected at baseline and liver biopsy at month 6; pairing these profiles under a donor ID without the visit term changes the biological estimand. Pairing a patient stool sample with a liver sample from a different patient is worse: it creates entirely fabricated association structure.

Donor	Stool visit	Plasma visit	Liver RNA visit	Eligible for same-visit model?
D001	T0	T0	T0	Yes
D002	T0	T0	T6	No; liver is six months later
D003	T0	missing	T0	No; one view absent
D004	T0	T0	T0	Yes
D005	T6	T6	T6	Yes, but time must enter analysis
D006	T0	T0	T0	Yes

A reproducible project should publish a matching report showing counts by individual assay, the complete-case intersection, partial-availability patterns, excluded samples with reasons, and whether paired or repeated-measures models were used. For small data, it is often better to explicitly declare a complete-case exploratory analysis and list its selection limitations than to “fill in” an unavailable omics view without a model-specific justification.

R audit A1 | Detect invisible identity mismatches

```
library(dplyr)
library(readr)
manifest <- read_csv("sample_manifest.csv",show_col_types=FALSE)
# One row per biological specimen/view at a specific visit.
req <- c("sample_id","subject_id","visit","assay","batch")
stopifnot(all(req %in% colnames(manifest)))
```

```
stopifnot(!anyNA(manifest[,req]))
stopifnot(!anyDuplicated(manifest$sample_id))
# Check whether one subject/visit has replicate library records.
ambiguous <- manifest %>% count(subject_id,visit,assay) %>%
            filter(n>1)
print(ambiguous)
# A complete subject-visit row requires all three layer names.
availability <- manifest %>% distinct(subject_id,visit,assay) %>%
                mutate(present=TRUE) %>%
                tidyr::pivot_wider(names_from=assay,values_from=present,
                                   values_fill=FALSE)
write_csv(availability,"omics_availability_by_visit.csv")
# Do not merge a T0 metabolome to a T6 transcriptome by subject_id.
# Follow-up action: audit many-to-many joins using declared relationship.
# Repeated-visits models split by subject_id, not specimen row.
```

Check your reasoning. If three distinct samples originate from one donor at three visits, are they independent in fivefold CV? No: their underlying donor-specific biology can leak into both training and test folds. If a single liver donor contributes 8,000 cells, are these 8,000 biological replicates for case/control inference? No: they are technical/sampling units within one donor. Cell-type pseudobulk aggregates cells by donor and cell type, then performs differential expression using donor-level replication.

Activity A. Ask your AI assistant to construct a toy manifest with six donors and three assays, deliberately including one wrong visit and one duplicate sample ID. In the generated analysis, identify exactly which assertion catches each error. State how many biological donors, not files, remain after valid matching. Your explanation must be written before you run the generated code.

Appendix B | What normalization does—and does not do

A “standardized” matrix is not automatically biologically comparable. Consider RNA counts, shotgun microbial taxa, quantitative proteomics and LC–MS peak areas. All can be converted to feature-wise z-scores, but those z-scores are dimensionless comparisons within each feature; they do not assert that a one-SD RNA change represents the same biochemical magnitude as a one-SD bile-acid change. Z-scoring eliminates scale differences but not differential detection limits, strong batch effects, inappropriate compositional treatment or selection bias.

Layer	Example input	Reasonable preprocessing	Key interpretation hazard
RNA-seq bulk	raw integer counts	Low-count filter + DESeq2 / edgeR modeling; VST for exploration	Never fit counts with a Gaussian model without transformation
scRNA pseudobulk	donor × cell-type sum counts	Donor-aware DE per cell type	Cell count per donor varies; avoid n=cells

Proteomics	intensity / peptide evidence	QC + missingness audit + log transform	MNAR censored proteins are not zeros
Targeted metabolomics	calibrated concentration	Standard/blank QC, dilution, batch correction	Absolute and relative quantification differ
Untargeted LC–MS	feature area + MS/MS ID	QC-based filtering, internal standards, ID grades	Unknown features cannot always map to KEGG
Microbiome composition	taxon counts / proportions	Prevalence filter, CLR/ALR or compositional models	Pseudocount and zero choices alter geometry

Microbiome-specific caution. Relative-abundance data obey a unit-sum constraint: an organism's apparent proportion can increase even when its absolute cell count remains stable and other taxa decline. CLR (centered log ratio) uses log ratios to a geometric mean and requires zero handling. For differential abundance, a compositional-aware method and careful simulation or absolute-load control may be more appropriate than unadjusted correlation of relative abundances. Disease, antibiotics and dietary exposures may influence both microbial community structure and bile-acid profiles.

R audit B1 | CLR geometry and zero handling

```
# Teaching: CLR for a taxon x sample count matrix.
# This is only one preprocessing option, not a universal microbiome model.
X <- as.matrix(read.csv("microbiome_taxa_counts.csv",
                        row.names=1,check.names=FALSE))
stopifnot(all(is.finite(X)),all(X>=0),all(colSums(X)>0))
prev <- rowMeans(X>0)
X <- X[prev>=0.20,,drop=FALSE]
stopifnot(nrow(X)>1)
clr <- function(z,pseudocount=0.5) {
  L <- log(z+pseudocount)
  sweep(L,2,colMeans(L),FUN="-")
}
Y <- clr(X)
write.csv(Y,"microbiome_clr.csv")
# Sensitivity analysis: repeat with a justified alternative
# zero strategy; check whether major associations persist.
# Important: unscaled CLR output still needs feature QC for MOFA2.
```

Sensitivity checks to request from an agent. Compare biological conclusions before and after batch correction; test alternative feature-prevalence thresholds; compare nominal associations to BH-adjusted discoveries; inspect whether top correlations survive antibiotic-use adjustment; retain a

table of all features removed and why. Changes across thresholds should be reported as instability, not hidden by selecting the most attractive setting.

Appendix C | ORA and GSEA: a worked numerical thought experiment

Suppose 15,000 genes passed expression filters. A preregistered threshold identifies 400 differentially expressed genes, and a KEGG pathway contains 100 of the tested genes; 15 pathway members are significant. A valid 2 × 2 table compares 15 versus 85 pathway genes and 385 versus 14,515 non-pathway genes. Under the null of uniform selection, the expected selected pathway members are 100 × 400 / 15,000 = 2.67, well below 15. Yet this calculation addresses only over-representation; it does not say which enzymes are active, the pathway direction, or whether the effect persists after annotation changes.

	Selected DE genes	Not selected	Total
In pathway	15	85	100
Outside pathway	385	14,515	14,900
Total	400	14,600	15,000

R audit C1 | Test an ORA contingency table

```
# Toy calculation; NOT output from a real MASLD experiment.
N <- 15000L; K <- 100L; n <- 400L; k <- 15L
expected <- n*K/N
p_over <- phyper(q=k-1,m=K,n=N-K,k=n,lower.tail=FALSE)
print(c(expected_overlap=expected,one_sided_p=p_over))
# This one pathway p-value is not the multiple-testing FDR.
# Across all pathway tests, use p.adjust(pvalues,method="BH").
```

Now imagine two gene sets, A and B. Set A contains a small group of strong DE genes just above the threshold, whereas set B contains many modest but consistent positive Wald statistics that do not individually meet |log2FC| > 1. ORA may detect A and miss B; ranked GSEA may detect B if its genes accumulate toward the positive tail. Neither method is inherently superior. They answer distinct questions and respond differently to gene correlation, set size, length/abundance biases, the tested universe and the direction of the ranking statistic.

A ranked statistic must be explicitly tied to its contrast. If the DESeq2 design reports log2FC(MASLD / Control), positive Wald statistics reflect genes that tend to be greater in MASLD under that model; if the reference level is reversed, the NES sign changes. A “positive pathway” statement without the reference contrast is uninterpretable. For pseudobulk analyses, also record which cell type, donor count and contrast produced the ranking.

False discovery rate is not a probability that one pathway is true. An adjusted p-value controls an expected rate of false discoveries under assumptions across a family of hypotheses. Different GO ontologies or repeated exploratory choices define potentially different multiplicity structures. If the

primary analysis tested a prespecified set of pathways and exploratory analyses assessed thousands more, the distinction must be preserved in the report.

Activity C. Ask an AI assistant to generate two schematic ranked gene lists with the same ORA hit count but different signed ranks. Make it explain why GSEA differs, with a plain-English null hypothesis. Independently audit whether its ranking vector is complete, numerical, finite, descending, and free of duplicate names. Reject any answer that infers enzyme activity solely from pathway membership.

Appendix D | Building a cross-omics evidence ledger

A major limitation of pathway integration is annotation asymmetry. Gene pathways often contain dozens to hundreds of annotated components, while a metabolomics assay may quantify only a small handful of metabolites in the same pathway, sometimes without distinguishing chemical isomers. Likewise, 16S-based microbial functional predictions largely recapitulate information learned from reference genomes rather than directly measuring genes in the assayed sample. “Three layers support the same pathway” can conceal three radically different kinds of evidence.

Evidence source	Observed object	What the result can support	What it cannot establish
RNA GSEA	Signed gene rank + annotation	Coordinated expression tendency	Protein activation / metabolite flux
Proteomics	Measured protein intensities	Protein-abundance association	Catalytic rate without assay
Phosphoproteomics	Site-level abundance / occupancy	Regulatory-site association	Net signal direction without context
Targeted LC–MS	Validated compound concentration	Measured metabolite change	Origin / tissue of production by itself
Shotgun metagenomics	Microbial genes / KO potential	Genomic metabolic capacity	In situ activity / metabolite production
16S prediction	Reference-inferred KOs	Plausible functions to follow up	Measured gene counts or transcripts
Spatial RNA	Spot/cell location and transcripts	Localization and tissue neighborhood	Cell–cell communication as direct proof

Strong pathway concordance requires more than shared names. Verify that the same species, tissue context and pathway identifier are being considered. The pathway described as “bile acid metabolism” in a plasma metabolomics output may not be identical to a KEGG liver gene-expression map. Examine overlap genes and compounds, and annotate whether the link is metabolic reaction chemistry, a transporter, a transcriptional regulatory relationship, or merely a broad label. Record these differences as structured evidence fields.

A practical ranking of hypotheses can remain qualitative and auditable. Assign each candidate the following independent dimensions: technical quality (batch/QC), statistical robustness (FDR, effect uncertainty, sensitivity), biological specificity (key molecules or cell type), mapping confidence (validated IDs), consistency across modalities, and experimental tractability. Avoid collapsing these to a one-number "causal score" unless a validated statistical interpretation exists.

Example candidate	Supporting observations	Crucial alternatives	Discriminating experiment
Microbial modification of bile acids	Taxa / measured bile-acid shifts / liver rank	Diet, medication or liver disease changes both	Isotope tracing plus microbial perturbation
Host inflammatory pathway	Pseudobulk macrophage terms + proteins	Immune-cell proportion or fibrosis stage	Cell-type-resolved functional perturbation
Hepatic lipid handling	RNA transport terms + plasma lipids	Systemic insulin resistance	Hepatocyte perturbation and flux assay

Activity D. Choose any candidate. Draft a defensible two-sentence result and a one-sentence limitation. Then propose the least expensive experiment that can distinguish it from its strongest noncausal alternative. The final claim must specify the evidence layer(s), direction of the explicit contrast and whether a causal test has actually been performed.

Appendix E | What the most recent papers change in practice

2026 GO knowledgebase update [8]. Background: functional annotations are continuously revised, making fixed "biological truths" out of pathway labels problematic. What is new: expanded curation, human functionome updates and activity-causal GO-CAM models support more structured interpretation. Major message: enrichment reports should include ontology and annotation versions. Remaining challenge: gene annotations remain uneven across organisms, tissues and mechanisms, and an ontology model does not substitute for intervention.

2026 Reactome Knowledgebase [10]. Background: a gene-list result hides the hierarchy of curated reactions and participant roles. Update: richer pathway exploration and refreshed multi-omics capabilities broaden interpretation beyond isolated terms. Major message: look inside the pathway graph before claiming directionality. Remaining challenge: coverage and curation reflect the literature, and reactions are not direct measurements of flux in the experiment.

2025 single-cell integration benchmarks [15,16]. Background: rapidly proliferating multimodal models have been compared using inconsistent datasets and metrics. What is distinctive: extensive method/task combinations, paired versus unpaired versus mosaic data geometry and metric-specific evaluation. Major finding: methods can rank differently for clustering, batch correction, feature recovery or spatial registration. Challenge: benchmarks do not ensure transfer to a new tissue, assay chemistry, donor population or rare cellular state.

2026 SCMBench [17]. Background: foundation models increasingly claim broad single-cell applicability. Methods: evaluated 23 domain-specific and foundation-model approaches across multiple tasks including representation integration, biomarker analysis, trajectory inference and batch correction. Major message: pretraining alone does not create a universal winner, and task-appropriate evaluation is essential. Challenge: generalization, biological feature fidelity and new-platform robustness require continued independent benchmarking.

2026 practical MOFA/DIABLO protocol [21]. Background: unsupervised and supervised tools are often described conceptually without sufficient preparation and interpretation guidance. Unique contribution: a hands-on integrated protocol covering data assembly, model construction and analysis outputs. Major practical lesson: a workflow must be connected to its study goal and quality checks. Challenge: small-n predictive validation, clinical transportability and causal inference still require an independent study design.

2024–2026 metabolomics updates [7,20]. Background: metabolomics begins with feature detection, identification confidence, quantitative QC and missingness, not just a pathway table. New protocols expand integrated analyses and interpretation possibilities, including exposure-related investigations. Major lesson: compound-identity and concentration evidence determine which multi-omics pathway claims are credible. Challenge: chemically unresolved features, shared metabolites across pathways and varying analytical coverage remain limiting.

Journal-club assignment. For one of [15–17], identify (a) the benchmark's evaluation unit and dataset split, (b) whether the authors test cell-type recovery, biological feature fidelity and batch removal separately, and (c) how you would adapt the same benchmark design to a gut–liver cohort. Summarize in 150 words, including one weakness and one concrete validation proposal.

Appendix F | A reusable acceptance contract for AI-generated analyses

Rather than asking a generative system to "do multi-omics integration and discover mechanisms," specify the input and evidence contract below. Such a contract reduces hidden assumptions while preserving student control of the scientific question. It can be used in ChatGPT, an agent-enabled IDE, or an R project using generated scripts.

AI-agent acceptance contract | Copy and adapt to your own dataset

```
# Example analysis contract (human-reviewed before execution)
project: week9_masld_tutorial
question: "Which pathway-level associations converge across layers?"
independent_unit: "subject at prespecified visit"
inputs:
  manifest: "sample_manifest.csv"
  transcriptome: "pseudobulk_DE_results.csv"
  tested_universe: "tested_gene_universe.txt"
  metabolites: "metabolite_pathway_results.csv"
required_checks:
  - "No duplicate or cross-visit sample IDs"
  - "Report samples retained and excluded by layer"
```

```
  - "Record organisms, feature/compound ID mapping rates"
  - "Use a signed all-gene ranking for GSEA"
  - "Use tested-gene background for ORA"
  - "Use BH correction for a named hypothesis family"
  - "Never merge noncommensurate quantitative scales"
  - "Disclose all imputation, filtering and batch choices"
outputs:
  - "matching_audit.csv"
  - "gene_identifier_mapping.csv"
  - "GO_BP_GSEA.csv"
  - "KEGG_GSEA.csv"
  - "multiomics_pathway_evidence.csv"
  - "R_sessionInfo.txt"
  - "interpretation_limitations.md"
stop_if:
  - "IDs or contrast are ambiguous"
  - "Matched sample count does not meet the planned design"
  - "Reference/source DOI could not be verified"
  - "Any output is described as causal without a causal test"
```

Final self-check before submitting. Can another student reproduce your figures from the same named inputs and recorded package/database versions? Can they find the donor/sample exclusions, mapping losses, uncertainty estimates and the intended statistical comparison? Do they know which statements represent direct measurement, inference from annotation, literature support or actual experimental causation? If any of these answers is “no,” the analysis is not yet ready for a mini-paper.